

Trainings ▾

Preparatory Steps

Automotive with CM870

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulators.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



- Drain the engine coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>)
- Remove the water pump drive belt. Refer to Procedure 008-003 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-003.html>)
- On some applications, removal of the alternator is required. Refer to Procedure 013-001 in Section 13. (</qs3/pubsys2/xml/en/procedures/10/10-013-001-tr.html>)
- Remove the air intake pipe to the turbocharger. Refer to the original equipment manufacturer (OEM) service manual.
- Remove the heat shield. Refer to Procedure 011-032 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-032-tr.html>)
- Disconnect the exhaust gas recirculation (EGR) valve motor and position sensor connectors from the harness.
- Remove the EGR valve coolant lines. Refer to Procedure 011-030 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-030-tr.html>)

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Note : Brush away all loose dirt from around the area of the air handling connections to avoid contamination of the interior of the engine.

- Disconnect the EGR temperature sensor. See the following procedure in the ISX CM871 and ISM CM876 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4021560. Refer to Procedure 019-378 in Section 19. (</qs3/pubsys2/xml/en/procedures/101/101-019-378.html>)
- Remove the front EGR connection tube from the EGR valve. Refer to Procedure 011-025 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-025-tr.html>)
- Disconnect the OEM fuel return hose from the integrated fuel system module. Refer to the OEM service manual. Or, loosen the fuel hose enough to position it away from the EGR valve mounting capscrew and tighten the fuel line.
- Remove the EGR valve heat shield. Refer to Procedure 011-032 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-032-tr.html>)
- Disconnect the OEM air compressor discharge line. Refer to the OEM service manual.

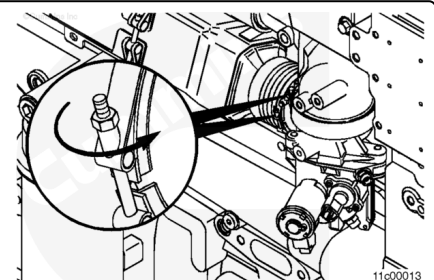


Remove

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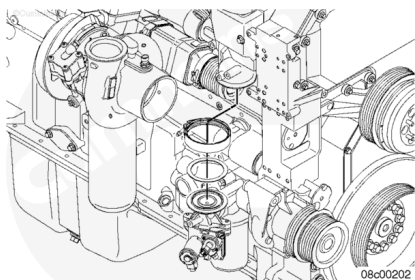
Loosen the V-band clamp connecting the EGR valve to the exhaust manifold.

Loosen the V-band clamp connecting the EGR valve to the EGR cooler connection.



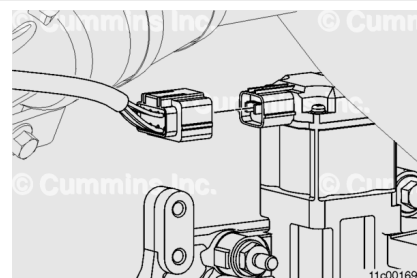
Remove the EGR valve and V-band clamps.

Remove and discard the gaskets.

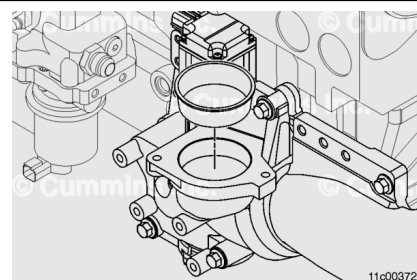


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Disconnect the EGR valve electrical connector.



Make sure to support the EGR valve by hand before removing the mounting capscrews. If the EGR valve is **not** supported, it can fall off the engine, causing damage to the EGR valve and actuator.

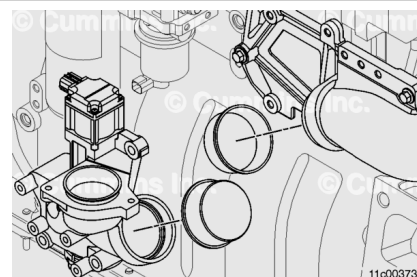


Loosen and remove the four EGR valve mounting capscrews.

Remove the EGR valve from the EGR valve support bracket.

The EGR valve will need to be moved at an angle forward and downward in order to disengage it from the EGR valve support bracket.

Use protective caps from the Air Handling Clean Care Kit to cover open points on the EGR valve and bracket.



Disassemble

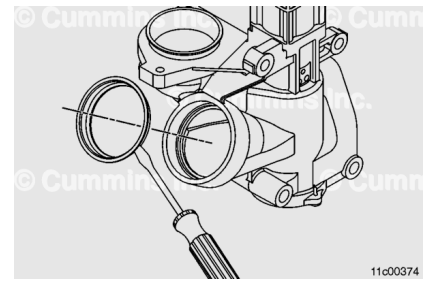
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Note : Be careful not to nick the EGR valve housing when removing the seal.

Remove the pipe seal from the EGR valve inlet port.

Pry the lip of the pipe seal up from the EGR valve with a small flat-headed screwdriver.

Discard the pipe seal.



Clean and Inspect for Reuse

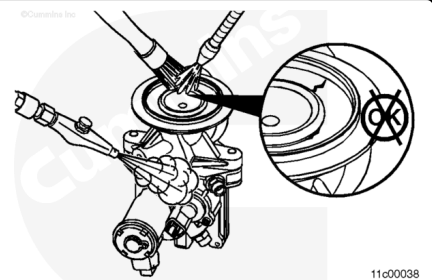
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⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of injury.



Check to make sure the EGR poppet head is present. If the poppet head is missing, run a stiff wire through every port of the exhaust manifold to locate the missing poppet head.

Do **not** assemble the engine without finding the missing poppet head.

Clean the mounting surfaces with a shop towel wet with a safety solvent to remove any deposits or debris.

Dry with compressed air.

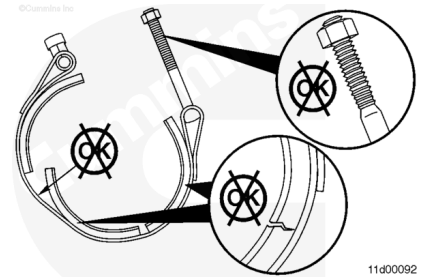
Inspect the mounting surfaces for cracks, fretting, or other damage. Inspect the EGR motor and position sensor for damage.

Replace the valve if damage is found.

Inspect the V-band clamps for signs of over-extension. The band **must not** be bent or cracked.

Inspect the V-band clamp threads for damage.

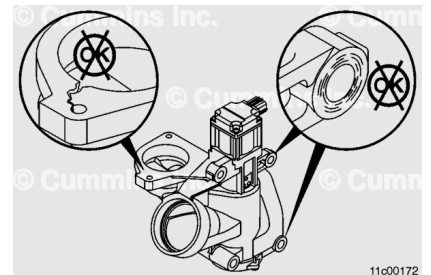
Replace the V-band clamp if damage is found.



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⚠ WARNING ⚠

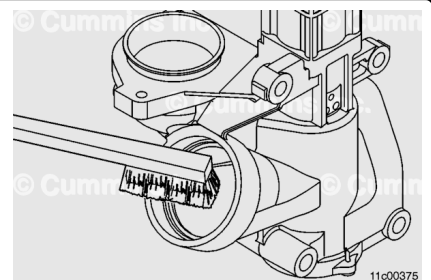
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the EGR valve mounting and sealing surfaces with a shop towel wet with safety solvent to remove any deposits or debris.

Clean the accumulated soot from the EGR valve inlet port using a wire brush, then soak with carburetor cleaner and clean again with a wire brush. Be sure the loose soot is removed from the EGR valve housing. This can be done by vacuuming the soot out of the EGR valve inlet port.

Dry with compressed air.

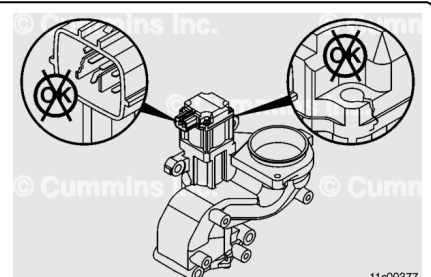
Inspect the mounting surfaces for cracks, fretting, or other damage.



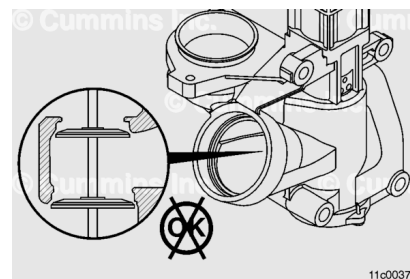
Inspect the EGR valve actuator for cracks or other damage. Replace if damage is found.

Inspect the EGR valve actuator connector for corrosion, worn pins, or other damage.

Replace the EGR valve assembly if corrosion, worn pins, or other damage is found.



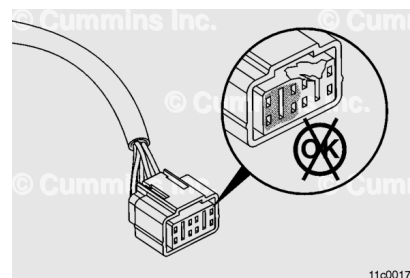
Inspect the EGR valve poppets for a stuck open condition. If the EGR valve is stuck open, clean or replace the EGR valve. Do **not** try to pry the poppets open or closed.



11c00376

Inspect the engine wiring harness EGR valve actuator connector for corrosion, worn receptacles, or other damage.

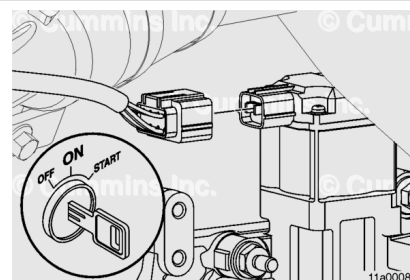
Replace the engine wiring harness EGR valve connector if corrosion, worn receptacles, or other damage is found.



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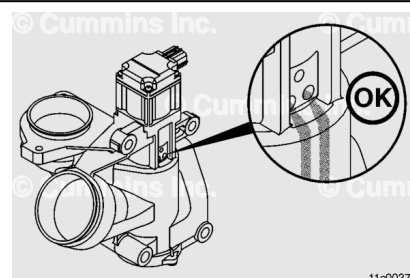
After cleaning the EGR valve, connect the electrical connector to the EGR valve and turn the ignition key ON. Do **not** start the engine.

Check for active Fault Code 1896. If Fault Code 1896 persists after cleaning, replace the EGR valve.



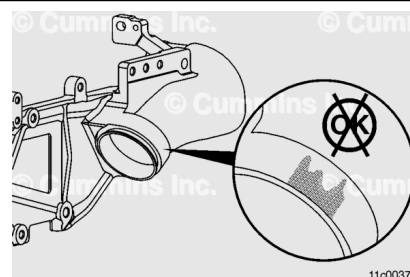
11a00085

Soot leakage at the EGR valve weep hole is normal and is to be expected. Do **not** replace the EGR valve if soot streaks are found at the EGR valve weep hole.



11c00378

Inspect the EGR valve mounting bracket for soot leaks at the EGR valve joint. If a leak is found, replace the EGR valve seal.

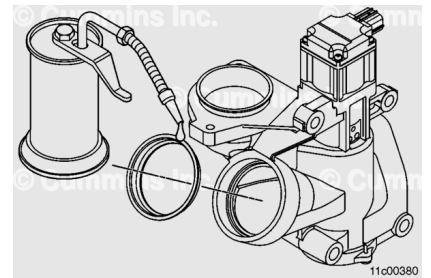


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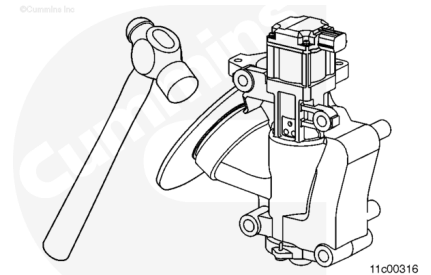
Assemble

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Use a light coat of clean engine oil to lubricate the outer diameter of the EGR valve seal during assembly onto the EGR valve. Push the EGR valve seal into the EGR valve inlet port by hand. The EGR seal **must** remain square in the EGR valve inlet port during installation.



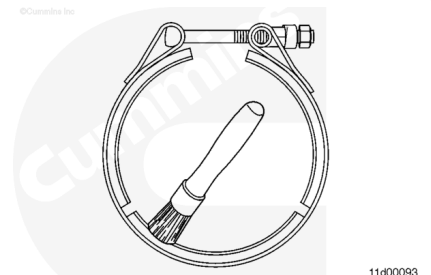
Hammer the EGR valve seal fully; use a flat piece of steel across the EGR valve seal. The EGR valve seal **must** be fully seated onto the EGR valve inlet port shoulder.



Install

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Before installing the V-band clamp, coat the clamping surface with high temperature anti-seize compound.

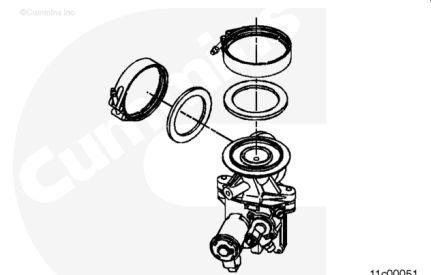


Install the smaller V-band clamp on the EGR cooler connection. Do **not** tighten.

Install the larger V-band clamp on the end of the EGR valve that connects with the exhaust manifold. Do **not** tighten.

Install a new gasket on the piloted end of the EGR cooler connection.

Install a new gasket on the piloted end of the EGR valve where it connects to the exhaust manifold.



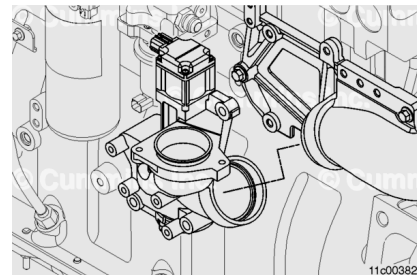
Install the EGR valve. Position the EGR connection V-band clamp, as shown, and tighten.

Torque Value: 17 n•m [150 in-lb]

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Lubricate the EGR valve seal with clean engine oil.

Install the EGR valve onto the EGR valve mounting bracket.

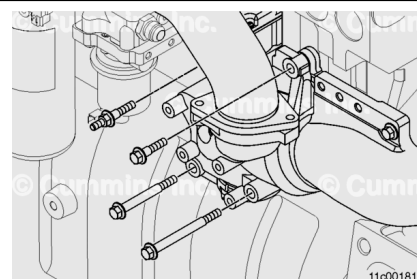


Install the four mounting capscrews into the EGR valve mounting bracket.

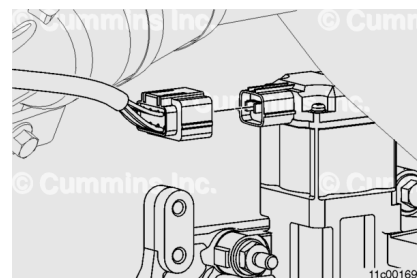
Tighten the capscrews.

Torque Value: 54 n•m [40 ft-lb]

Note : The longest capscrew passes through the EGR valve and EGR valve mounting bracket and threads into the engine block.



Connect the EGR valve connector.



Finishing Steps

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- Install the EGR valve coolant lines. Refer to Procedure 011-030 in Section 11.
(</qs3/pubsys2/xml/en/procedures/10/10-011-030-tr.html>)
- Connect the EGR valve motor and position sensor connectors to the harness (clean off the wiring harness



and connector before connecting to the valve).

- Install the heat shield. Refer to Procedure 011-032 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-032-tr.html>)
- Install the air intake pipe to the turbocharger. Refer to the OEM service manual.
- Install the alternator, if removed. Refer to Procedure 013-001 in Section 13. (</qs3/pubsys2/xml/en/procedures/10/10-013-001-tr.html>)
- Install the water pump drive belt. Refer to Procedure 008-003 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-003.html>)
- Fill the engine cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>)
- Operate the engine and check for exhaust and coolant leaks.

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- Install the EGR heat shield. Refer to Procedure 011-032 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-032-tr.html>)
- Connect the OEM air compressor discharge line. Refer to the OEM service manual.
- Install the front EGR connection tube. Refer to Procedure 011-025 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-025-tr.html>)
- Connect the OEM fuel return hose from the integrated fuel system module. Refer to the OEM service manual.
- Connect the EGR temperature sensor. See the following procedure in the ISX CM871 and ISM CM876 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4021560. Refer to Procedure 019-378 in Section 19. (</qs3/pubsys2/xml/en/procedures/101/101-019-378.html>)
- Operate the engine and check for exhaust or coolant leaks.
- Verify active fault codes. If a malfunction resulted in coolant, oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to Procedure 014-013 in Section



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14. (/qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html)

Last Modified: 13-May-2013
